BEFORE AND AFTER ANALYSIS

Capstone 23: AI Responsible Electronics — Standards, Safety and E-Waste Research

1. Before Using AI and Verification

At the beginning of this project, my understanding of India's E-Waste (Management) Rules, 2022 was limited. I knew that electronic waste should be recycled, but I did not fully understand Extended Producer Responsibility (EPR), registration requirements, or the responsibilities of producers and recyclers. Some AI-generated explanations were broad and could have been misleading if accepted without checking official sources.

2. Verification Process

To improve accuracy, I compared AI-generated information with official sources from the Ministry of Environment, Forest and Climate Change (MoEFCC) and the Central Pollution Control Board (CPCB). I checked key claims about the effective date of the rules, EPR, registration, the treatment of waste batteries under separate rules, and later amendments. I recorded the findings in a verification table and used a source-verification spreadsheet to track claims and their status.

3. After Verification

After verification, the project became more accurate and better supported by evidence. Broad or incorrect statements were revised, official references were added, and major claims were checked against government sources. The project materials included an awareness article, poster text, social media posts, a presentation outline, a prompt library, and a verification report.

4. Key Improvements

Before	After
General understanding of e-waste	Clearer understanding of the scope and key provisions of India's E-Waste Rules, 2022
Relied on AI summaries	Checked important claims against MoEFCC and CPCB sources
Some broad or potentially misleading claims	Revised claims and added supporting references
No consistent evidence tracking	Created a verification table and spreadsheet-based tracking workflow
Basic awareness material	Prepared research and awareness materials, prompts, and a verification report

5. Limitations of AI

AI can summarize information quickly, but it may miss recent amendments, confuse different environmental rules, or make claims that are too general. AI-generated information should therefore be treated as a starting point, not as legal advice. Important legal and technical claims should be checked against current official documents.

6. Conclusion

This project demonstrated that AI is useful for organizing research and drafting awareness content, but human judgment and source verification remain essential. Combining AI assistance with official government references helped produce a more responsible and evidence-based project on India's E-Waste (Management) Rules, 2022.

Official reference websites:
MoEFCC Rules and Regulations: https://moef.gov.in/rules-regulations-3
CPCB E-Waste Management System: https://eprewaste.cpcb.gov.in/